

Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

Work Experience

Perfomax

May 2026–present

AI Advisor

- **AI Strategy & Platform Architecture :**

- Advised on AI/ML strategy, product implementation, and technical design for the platform.

National University of Singapore

2022–present

Postdoc under Kostya Novoselov

- **Leadership :**

- Speaker search and selection for the 500-person AI4X 2025 conference.
- Main organizer of the ICLR 2025 workshop on multiscale machine learning.

- **Transformer for generating symmetric crystals :**

- Created a generative model for material design based on symmetry inductive bias.
- Implemented the model in PyTorch.
- Achieved the best generation quality and diversity among space-group-conditioned models.
- Led a team of 6.

- **ML for predicting properties of defects in 2D crystals :**

- Created a sparse representation of crystals with defects, achieving 4x energy prediction error reduction.
- Developed code for reproducible parallel running of machine learning experiments.
- Led a team of 3.

Constructor Group

February 2021–present

Machine Learning Consultant

- **AI/ML Consulting :**

- Consulted on machine learning algorithms, deep learning applications, and architectural designs for scientific and industry-led projects.

CERN [Yandex → HSE University]

2014–2022

Researcher

[2025 Breakthrough Prize in Fundamental Physics](#) as a member of LHCb.

- **Muon identification at the LHCb experiment at CERN :**

- Solved a classification problem over tabular data with noisy labels under timing constraints.
- Developed a gradient boosting model that reduced the false positive rate by 30% in the critical low-track-momentum region.
- Integrated the model into LHCb production, mostly in C++.
- Packaged the work as a data science competition problem for IDAO-2019.

- **CatBoost aka fighting biases with dynamic boosting :**

- Set up a distributed system for running experiments for the team with Bash and Python.
- Studied gradient boosting improvements with experiments on toy data.

Education

Higher School of Economics (HSE University)

2016–2020

PhD in Computer Science, supervisor [Andrey Ustyuzhanin](#); [Machine Learning for particle identification in the LHCb detector](#).

Sapienza — Università di Roma

2016–2020

PhD in Physics (double degree with HSE), supervisor [Barbara Sciascia](#).

Product Management course at Yandex

Spring 2018

Focused coursework in product strategy and execution.

Yandex School of Data Analysis

2013–2015

Master's level CS course covering algorithms, machine learning, deep learning, and distributed systems.

Moscow Institute of Physics and Technology

MS in Physics (2014–2016): Optimisation of data processing of the LHCb experiment.

BS in Physics (2010–2014): Study of the quantum states of the electrons in nonideal plasma and selected molecules using wave packet molecular dynamics with packet splitting.

Mentorship

- Mentored 9 students ([1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#)), 3 interns, and student workshop projects ([1](#), [20](#)).
- Student council member from 2013 to 2016, leading several dormitory-scale IT projects.
- Co-PI of a \$3.4m AI Singapore grant on multiscale machine learning.

Teaching & Outreach

- Pitch research to the general public through major newspaper coverage ([1](#), [2](#)), blog posts ([1](#), [2](#)), a [science slam](#), interviews ([1](#), [2](#)), and a public [talk](#).
- Taught at the Machine Learning in High Energy Physics Summer School in [2015](#), [2016](#), [2017](#), [2018](#), [2019](#), [2020](#), [2021](#), and [2023](#).
- Lecturer in the [Coursera Advanced Machine Learning specialisation](#), which is [award-winning](#).
- Taught and developed an introductory data analysis [course](#) at HSE.
- Delivered more than 20 [conference talks](#).

Service

- Reviewer for RSC Advances, Machine Learning: Science and Technology, and AI for Accelerated Materials Design workshops at NeurIPS and ICLR (2023–2025).
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

Skills

ML & AI (Structured, non-text, domain-specific data) ·
Python (Research coding) ·
PyTorch (Deep learning) ·
Linux (Administration) ·
Research Writing & Speaking ·
Leadership & Mentorship

Selected Publications

1. **Kazeev, Nikita**, and Phan Bui Nhat Huyen. [Position: Align AI to Our Aspirations, Not Our Flaws](#). Pluralistic Alignment Workshop at ICML 2026, June 2026.
2. Artem Dembitskiy, Shuya Yamazaki, Artem Maevskiy, **Nikita Kazeev**, Roman A. Eremin, Semen Budenny, Kedar Hippalgaonkar, Antonio Helio Castro Neto, and Andrey E. Ustyuzhanin. [Generative Pipeline for Discovering Solid-State Battery Materials with Universal Atomistic Potentials](#). ICML 2026 AI for Science Workshop, May 2026.
3. Jose Recatala-Gomez, **Nikita Kazeev**, et al. [Generative design of inorganic materials](#). arXiv preprint arXiv:2604.14082, April 2026.
4. Alexandre Duval, Siddharth Betala, Samuel P. Gleason, Andy Xu, Georgia Channing, Daniel Levy, Ali Ramlaoui, Clémentine Fourier, Chaitanya K. Joshi, **Nikita Kazeev**, Sékou-Oumar Kaba, Félix Therrien, Alex Hernández-García, Rocío Mercado, N M Anoop Krishnan. [LeMat-GenBench: Bridging the gap between crystal generation and materials discovery](#). AI for Accelerated Materials Design - NeurIPS 2025 / arXiv preprint arXiv:2512.04562, October 2025.
5. Stephen G. Dale, **Nikita Kazeev**, et al. [AI4X Roadmap: Artificial Intelligence for the advancement of scientific pursuit and its future directions](#). arXiv preprint arXiv:2511.20976, November 2025.
6. **Kazeev, Nikita**, et al. [WyckoffTransformer: Generation of Symmetric Crystals](#). ICML 2025, May 2025.
7. **Kazeev, N.**, Al-Maeeni, A.R., Romanov, I. et al. [Sparse representation for machine learning the properties of defects in 2D materials](#). npj Comput Mater 9, 113 (2023).